

Question 1: In a college canteen survey of **400 students**, the following data was collected:

- 180 students prefer **Cricket**
- 220 students prefer **Football**
- 120 students prefer **both Cricket and Football**

Using this information, answer the following:

1. Calculate the **support** of the itemset {Cricket, Football}.
2. Calculate the **confidence** of the rule:
{Cricket}→{Football}
3. Calculate the **confidence** of the rule:
{Football}→{Cricket}
4. Calculate the **lift** of the rule:
{Cricket}→{Football}
5. Interpret whether Cricket and Football preferences are positively associated, negatively associated, or independent.

Question 2: Consider the following transaction database:

TID	Items Purchased
T1	{Pen, Notebook, Pencil}
T2	{Notebook, Pencil}
T3	{Pen, Eraser}
T4	{Pen, Notebook}
T5	{Notebook, Pencil, Eraser}
T6	{Pen, Notebook, Pencil}

Answer the following:

1. Calculate the **support count** and **support** of the itemset {Pen, Notebook}.
2. Calculate the **confidence** of the rule:
 $\{Pen\} \rightarrow \{Notebook\}$
3. Calculate the **confidence** of the rule:
 $\{Notebook\} \rightarrow \{Pen\}$
4. Calculate the **lift** of the rule:
 $\{Pen\} \rightarrow \{Notebook\}$
5. If minimum support = 50% and minimum confidence = 60%, determine whether the rule
 $\{Pen\} \rightarrow \{Notebook\}$ is a **strong association rule**.

Question 3: A medical diagnostic model was tested on 200 patients for detecting a disease. The results are given below:

- 80 patients actually had the disease.
- Out of these, 70 were correctly predicted as positive.
- 20 patients were wrongly predicted as positive.
- The remaining patients were correctly predicted as negative.

Answer the following:

1. Construct the Confusion Matrix.
2. Calculate:
 - Accuracy
 - Precision
 - Recall (Sensitivity)
 - Specificity
 - F1-Score
3. Interpret whether the model is better at identifying positive cases or negative cases.

Question 4: A machine learning model was developed to predict whether a student will Pass or Fail an exam. After testing the model on 150 students, the following Confusion Matrix was obtained:

	Predicted Pass	Predicted Fail
Actual Pass	60	15
Actual Fail	10	65

Using the above confusion matrix, answer the following:

1. Identify the values of:
 - True Positive (TP)
 - True Negative (TN)
 - False Positive (FP)
 - False Negative (FN)
2. Calculate the following performance metrics:
 - Accuracy
 - Precision
 - Recall (Sensitivity)
 - Specificity
 - F1-Score
3. Based on the results, comment on whether the model is better at predicting **Pass** or **Fail** cases.

Question 5: Let a random variable XXX take the following values:

X	Probability
1	0.2
3	0.5
5	0.3

Let $\phi(x)=x^2$.

1. Compute $E[X]$.
2. Compute $\phi(E[X])$
3. Compute $E[\phi(X)]$
4. Verify Jensen's Inequality numerically.

Question 5: A discrete random variable XXX takes the values 1, 3, and 5 with probabilities 0.2, 0.5, and 0.3, respectively. Let the function be $\phi(x)=x^2$, which is a convex function.

Calculate the expected value $E[X]$, then compute $\phi(E[X])$.

Next, calculate $E[\phi(X)]=E[X^2]$. Finally, compare the values of $\phi(E[X])$ and $E[\phi(X)]$ and verify numerically whether Jensen's Inequality holds in this case.

Question 6: A retail company collects data on customers including annual income, spending score, and number of monthly purchases to understand buying behavior. The management applies Principal Component Analysis (PCA) to reduce dimensionality and finds that the first principal component explains 72% of the total variance. To further validate whether customer spending behavior has significantly increased after a new marketing campaign, the company selects a random sample of 30 customers and records an average spending increase of ₹1,500 with a known population standard deviation of ₹3,000. At a 5% significance level, determine how many principal components should be retained if at least 80% variance must be preserved, and perform an appropriate hypothesis test to check whether the marketing campaign significantly increased customer spending.

Question 7: A dataset contains the monthly sales figures (in units) of a retail store as follows: 2450, 3800, 1250, 9100, 5600, and 4300. Using the decimal scaling normalization technique, normalize the data so that all values lie between -1 and 1 . Clearly determine the value of jjj used for scaling and compute the normalized values for each observation.

Question 8: A dataset contains the test scores of 8 students in a statistics exam as follows: 55, 60, 72, 68, 75, 80, 62, and 70. Using the Z-score normalization technique, standardize the data. First calculate the mean and standard deviation of the dataset, and then compute the Z-score for each observation. Interpret what a negative and a positive Z-score indicate in this context.

Question 9: A dataset contains the daily sales (in units) of a store for one week as follows: 120, 150, 200, 250, 300, 180, and 220. Using the Min-Max normalization technique, transform the data into a new range of $[0, 1]$. Clearly identify the minimum and maximum values, apply the normalization formula to each observation, and compute the normalized values.

Further, normalize the same dataset into the range $[10, 50]$ and compare the transformed results.

Question 10: A retail company maintains the following customer dataset:

Customer_ID	Age	Monthly_Income (₹)	Spending_Score
C1	25	40000	70
C2	30	—	65
C3	—	52000	80
C4	28	48000	—
C5	35	60000	75

Identify the missing values in the dataset.

Replace the missing values in:

- **Age** using mean imputation.
- **Monthly_Income** using median imputation.
- **Spending_Score** using mean imputation.

Show the updated dataset after handling missing values.

Briefly explain one disadvantage of mean imputation.